Classification and Measurement of One-Dimensional Hyperuniform Point Patterns

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April 20, 2026

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This paper represents my own work in accordance with University regulations.
/s/ Michael Fang

Abstract

For an uncorrelated random point pattern in space, the variance in the number of points found inside a spherical observation window of radius R grows in proportion to R^d in d dimensions. Hyperuniform patterns break this scaling, as their density fluctuations are anomalously suppressed at large scales. When the structure factor vanishes as $S(k) \sim |k|^\alpha$ near the origin, the exponent α defines three distinct hyperuniformity classes. However, patterns within the same class can have very different fluctuation amplitudes, motivating the use of a surface-area coefficient $\bar{\Lambda}$ for Class I patterns to measure the amplitude of the remaining fluctuations. In this work, we compute both α and $\bar{\Lambda}$ across a range of one-dimensional hyperuniform point patterns, including metallic-mean substitution chains whose irrational tile-length ratios generate quasiperiodic order, higher-rank substitution tilings, uniformly randomized lattices, and stealthy configurations. Starting from the eigenvalue formula for substitution matrices, we confirm that all metallic-mean chains share $\alpha = 3$ and find that $\bar{\Lambda}$ increases monotonically from Fibonacci (the golden ratio) toward an apparent limit near 1/3. We then identify algebraic constraints on the substitution matrix and show how relaxing these constraints opens up new regions of the α spectrum. Across all systems studied, patterns sharing the same α were determined to be able to have very different $\bar{\Lambda}$, showing that α and $\bar{\Lambda}$ capture complementary and independent information about large-scale structure.

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1 Introduction

Large-scale density fluctuations contain fundamental information about the structure of many-body systems. In particular, characterizing the local number variance of point patterns has widespread applications across condensed matter physics and beyond. From quasicrystals that exhibit sharp diffraction with forbidden rotational symmetries [1] to the photoreceptor mosaics of bird retinas [2] and the distribution of prime numbers on the real line [3], the large-scale density fluctuations are anomalously suppressed below what simple randomness would predict. The property that unites these systems is hyperuniformity.

For any given point pattern, we can consider the variance in the number of points in some arbitrarily-centered local observation window. For a one-dimensional random (Poisson) distribution, this variance grows linearly in proportion to the region's size, according to $\sigma^2(R) = 2\rho R$. This linear growth is characteristic of uncorrelated matter and is known as the Poisson baseline [4]. In a hyperuniform system, the variance will instead grow more slowly than the window size, even though the pattern may appear equally disordered at short distances [4, 5]. A point pattern is hyperuniform if and only if its structure factor $S(k)$ vanishes as $k \to 0$ [4].

The rate at which $S(k)$ vanishes determines the strength of hyperuniformity. Generally, hyperuniform patterns are sorted into three main classes depending on how quickly $S(k) \to 0$ as $k \to 0$, assuming $S(k) \sim |k|^\alpha$. Here, α determines whether the number variance is bounded (Class I, $\alpha > 1$), grows logarithmically (Class II, $\alpha = 1$), or grows as a weak power law (Class III, $0 < \alpha < 1$) [4, 5]. While this classification scheme is useful, it is still not able to fully distinguish between hyperuniform patterns. For example, two Class I patterns can both have $\alpha = 3$, but differ substantially in how much their density fluctuates at any given scale.

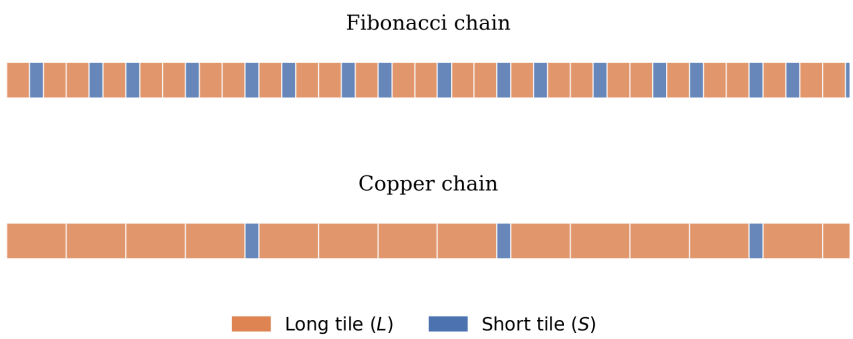

Figure 1: Two Class I hyperuniform chains, both with $\alpha = 3$, but with structurally distinct tile arrangements.

To distinguish between patterns within the same class, we need a metric that captures the amplitude of the fluctuations, not just their scaling rate. The surface-area coefficient $\bar{\Lambda}$, defined as the running average of the number variance at large R [4], serves this role. In this work, we compute both α and $\bar{\Lambda}$ for a broad collection of one-dimensional hyperuniform point patterns, including substitution tilings, perturbed lattices, and stealthy configurations. We identify the algebraic constraints that determine which α values are accessible, and we use $\bar{\Lambda}$ to rank patterns within each class. Across all systems studied, we find that α and $\bar{\Lambda}$ capture independent information, as patterns sharing the same α can differ significantly in $\bar{\Lambda}$.

2 Theory

2.1 Number variance, structure factor, and the hyperuniformity exponent α

Let us start by considering a one-dimensional point pattern with number density ρ. If we slide a window with half-width R along the pattern and count the number of points inside, the count will fluctuate from one window placement to the next. The variance of this count is the number variance, $\sigma^2(R)$.

For a Poisson process, each part of the system is statistically independent of the rest, so bigger windows accumulate more noise and the variance grows linearly ($\sigma^2_{\text{Poisson}}(R) = 2\rho R$) [4]. When points are spatially correlated, however, the variance picks up a correction that depends on the pair correlation function $g_2(r)$, defined as the ratio of the two-particle density to ρ^2 [4]. When $g_2(r) > 1$, points are more likely to be found at separation r than in an uncorrelated system. When $g_2(r) < 1$, they are less likely. The variance is [4]

$$\sigma^2(R) = 2\rho R \left[1 + 2\rho \int_0^{2R} \left(1 - \frac{r}{2R}\right) \left[g_2(r) - 1\right] dr\right]. \tag{1}$$

To connect the variance to the small-k behavior of the pattern, it is useful to work in Fourier space with the structure factor [4]

$$S(k) = 1 + \rho \hat{h}(k), \tag{2}$$

where $\hat{h}(k)$ is the Fourier transform of the total correlation function $h(r) = g_2(r) - 1$. The structure factor is non-negative for all k and equals 1 everywhere for a Poisson process. The number variance can then be rewritten as a weighted integral over $S(k)$ [4, 6]:

$$\sigma^2(R) = \frac{\rho}{\pi} \int_0^\infty S(k) \, \frac{4\sin^2(kR)}{k^2} \, dk. \tag{3}$$

Because $4\sin^2(kR)/k^2$ is peaked near $k = 0$ for large R, the large-scale behavior of $\sigma^2(R)$ is controlled by the small-k behavior of $S(k)$. Thus, suppressing large-scale density fluctuations is equivalent to requiring $S(k)$ to vanish near the origin. A point pattern is therefore called **hyperuniform** if and only if [4]

$$\lim_{k \to 0} S(k) = 0. \tag{4}$$

The rate at which $S(k)$ vanishes near the origin controls the large-R scaling of $\sigma^2(R)$ and defines three hyperuniformity classes. When $S(k)$ vanishes as a power law,

$$S(k) \sim |k|^\alpha \quad \text{as } k \to 0, \tag{5}$$

substituting into Eq. (3) and taking the $R \to \infty$ limit yields three qualitatively different behaviors [4, 5]:

$$\sigma^2(R) \sim \begin{cases} \text{const} & \alpha > 1 \quad \text{(Class I)}, \\ \Lambda_{\text{II}} \ln R & \alpha = 1 \quad \text{(Class II)}, \\ \Lambda_{\text{III}} R^{1-\alpha} & 0 < \alpha < 1 \quad \text{(Class III)}. \end{cases} \tag{6}$$

Class I is the strongest form of hyperuniformity, as the variance remains bounded no matter how large the window gets. All perfect crystals and quasicrystals generated by Pisot substitution rules[1] belong to this class [7]. For Class II hyperuniformity, the variance grows infinitely with R,

[1] A Pisot number is an algebraic integer greater than 1 whose algebraic conjugates all have absolute value less than 1. The golden ratio $\tau = (1 + \sqrt{5})/2 \approx 1.618$, whose conjugate is $(1 - \sqrt{5})/2 \approx -0.618$, is the simplest example.

but only logarithmically. Class III is the weakest form of hyperuniformity, with an unbounded variance that grows more slowly than that of a Poisson distribution.

2.2 The surface-area coefficient $\bar{\Lambda}$

Knowing which class a pattern belongs to does not fully characterize it. Two Class I quasicrystals can both have $\alpha = 3$ but differ substantially in how much their density fluctuates inside a window of any given size. To see this difference, we need a quantity that captures the amplitude of the fluctuations, not just their scaling.

For Class I patterns in one dimension, the variance does not simply converge to a constant but typically oscillates about a mean level. This is especially true for quasicrystals with dense Bragg spectra, where the oscillations persist at all scales. In order to extract a well-defined number from this behavior, we define the surface-area coefficient as the running average

$$\bar{\Lambda} \equiv \lim_{L\to\infty} \frac{1}{L} \int_0^L \sigma^2(R)\, dR. \tag{7}$$

This averages out the oscillations and converges to the mean level of the plateau. Torquato and Stillinger proved that the one-dimensional integer lattice achieves $\bar{\Lambda} = 1/6$, which is the global minimum among all infinite one-dimensional hyperuniform patterns [4]. For aperiodic systems, however, almost no values are known.

The definition of $\bar{\Lambda}$ in Eq. (7) yields a finite value only for Class I patterns, because the variance is bounded. For Classes II and III, then, the variance grows without bound and $\bar{\Lambda}$ diverges. To rank patterns in these classes, we need analogous coefficients that extract the leading amplitude from the appropriate growth law. For Class II ($\alpha = 1$), the variance grows logarithmically, so we define

$$\Lambda_{\mathrm{II}} = \lim_{R\to\infty} \frac{\sigma^2(R)}{\ln R}, \tag{8}$$

which isolates the prefactor of the logarithmic growth.

For Class III ($0 < \alpha < 1$), the variance grows as a power law, so we define

$$\Lambda_{\mathrm{III}} = \lim_{R\to\infty} \frac{\sigma^2(R)}{R^{1-\alpha}}, \tag{9}$$

which isolates the prefactor of the power-law growth. The procedures used to extract Λ_{II} and Λ_{III} from finite data are described in Sec. 3.2.

2.3 Substitution tilings and the eigenvalue formula

Many of the patterns we study are generated by substitution rules. A one-dimensional substitution tiling starts with a short seed word over a finite alphabet and grows by repeatedly replacing each letter with a fixed string of letters. After many iterations, the result is a long aperiodic sequence. By assigning each letter a physical tile length and placing the tiles end to end, one can produce a one-dimensional point pattern at the tile boundaries.

The growth of the tiling is governed by the substitution matrix M, an $r \times r$ non-negative integer matrix (for r tile types) whose entry M_{ij} counts how many tiles of type i are produced when a single tile of type j is substituted [7]. For example, the Fibonacci chain uses two tile types, S (short) and L (long), with the rule $S \to L$, $L \to LS$. Its substitution matrix is

$$M_{\mathrm{Fib}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \tag{10}$$

with eigenvalues $\lambda_1 = \tau = (1+\sqrt{5})/2 \approx 1.618$ (the golden ratio) and $\lambda_2 = -1/\tau \approx -0.618$. The determinant $|\det M| = |0 \cdot 1 - 1 \cdot 1| = 1$, so $|\lambda_2| = 1/\lambda_1$. All metallic-mean matrices share this property, and as we show in Sec. 4.2, this constrains which values of α are accessible.

The largest eigenvalue λ_1 is always positive and real for non-negative integer matrices. Every time the substitution rule is applied, the total length of the tiling is multiplied by approximately λ_1. The corresponding eigenvector then gives the asymptotic relative frequencies of each tile type, and the tile lengths are set so that the pattern scales consistently. For the metallic-mean family, where $M = \left(\begin{smallmatrix} 0 & 1 \\ 1 & n \end{smallmatrix}\right)$, the tile lengths are fixed by requiring $L/S = \lambda_1 = \mu_n$ [7].

Substitution matrices and hyperuniformity are connected by the eigenvalue formula of Oğuz et al [8, 7]:

$$\alpha = 1 - \frac{2 \ln |\lambda_2|}{\ln \lambda_1}. \tag{11}$$

For substitution tilings, this equation determines α entirely from the two largest eigenvalues. Physically, λ_1 sets how much the tiling grows with each substitution step, while $|\lambda_2|$ controls how fast density fluctuations decay. A smaller $|\lambda_2|$ corresponds to stronger suppression of large-scale density fluctuations and a larger α. From Eq. (11), the tiling is hyperuniform ($\alpha > 0$) whenever $|\lambda_2| < \sqrt{\lambda_1}$.

When λ_1 is a Pisot number, meaning all other eigenvalues have absolute values less than 1 [7], the diffraction spectrum consists entirely of Bragg peaks, which are sharp delta-function peaks at a dense, countable set of wavevectors [7]. This distinguishes quasicrystals from periodic crystals, whose Bragg peaks are discrete and well-separated, and from disordered systems, whose $S(k)$ is smooth and continuous. Because the Bragg peaks are dense on the real line, one cannot extract α by fitting a power law to $S(k)$ directly. To overcome this, we use diffusion spreadability [9, 10], which smooths over the Bragg spectrum and provides a clean route to α.

2.4 Diffusion spreadability

To extract α from a point pattern, we convert it into a heterogeneous medium and study how transport through that medium reflects the pattern's large-scale structure. For a one-dimensional point pattern, we can construct a two-phase medium by centering a solid rod at each point, with the remaining space treated as void. The quantity that governs transport through this medium is not the bare structure factor $S(k)$, which describes correlations between points, but the spectral density

$$\tilde{\chi}_V(k) = \rho\, |\tilde{m}(k)|^2\, S(k), \tag{12}$$

where $\tilde{m}(k)$ is the Fourier transform of the rod indicator function [9]. The form factor $|\tilde{m}(k)|^2$ provides a smooth, decaying envelope that damps the Bragg peaks at large k, making $\tilde{\chi}_V(k)$ better behaved than $S(k)$ itself.

The diffusion spreadability $\mathcal{S}(t)$ measures how quickly solute, initially distributed uniformly in the solid phase and absent from the void, spreads into the void over time. In d dimensions, the excess spreadability has the exact Fourier-space representation [9]

$$\mathcal{S}(\infty) - \mathcal{S}(t) = \frac{1}{(2\pi)^d\, \phi_2} \int \tilde{\chi}_V(\mathbf{k})\, e^{-k^2 D t}\, d\mathbf{k}, \tag{13}$$

where ϕ_2 is the packing fraction and D is the diffusion coefficient. The Gaussian kernel $e^{-k^2 D t}$ acts as a progressive low-pass filter on the spectral density [10, 11]. At short times, all Fourier modes contribute and the dense Bragg peaks dominate the sum. As t grows, the kernel damps high-k modes first, progressively smoothing out the Bragg spectrum until only the small-k behavior of

$\tilde{\chi}_V(k)$ survives. This smoothing is what allows spreadability to extract a clean power law from a dense Bragg spectrum, where direct fitting of $S(k)$ cannot [10].

If the spectral density vanishes as $\tilde{\chi}_V(k) \sim |k|^\alpha$ near the origin, the excess spreadability decays at long times as [9]

$$\mathcal{S}(\infty) - \mathcal{S}(t) \sim t^{-(d+\alpha)/2}. \tag{14}$$

In one dimension ($d = 1$), this becomes $t^{-(1+\alpha)/2}$. The mechanism behind this decay is that larger α suppresses long-wavelength modes of $\tilde{\chi}_V(k)$ more strongly, so $\tilde{\chi}_V(k)$ vanishes more quickly near $k = 0$ and the solute spreads to uniformity faster. For example, a one-dimensional Class I pattern with $\alpha = 3$ has excess spreadability decaying as t^{-2}, while one with $\alpha = 1.5$ decays only as $t^{-1.25}$. The hyperuniformity exponent therefore connects the structure of a point pattern to measurable physical properties of the corresponding two-phase medium.

2.5 Other hyperuniform systems

The substitution tilings described in Sec. 2.3 are all deterministic and ordered, with α fixed by the substitution matrix. To place these tilings in a broader context, we also study two comparison families that achieve hyperuniformity through different mechanisms.

The uniformly randomized lattice (URL) is constructed by taking the one-dimensional integer lattice and independently displacing each site by a uniform random amount on $[-a/2, a/2]$ [12]. Because the displacements are independent, the only correlations between sites come from the underlying lattice. On large scales, any window of half-width $R \gg a$ contains close to $2\rho R$ points, with fluctuations only from the boundary region of width $\sim a$ where a displaced site may or may not fall inside. These boundary fluctuations do not grow with R, so $\sigma^2(R)$ is bounded and the URL is Class I hyperuniform with $S(k) \sim k^2$, giving $\alpha = 2$ for all $a > 0$. For URLs, an exact expression for $\bar{\Lambda}$ in one dimension [12] is

$$\bar{\Lambda}(a) = \frac{a}{3} + \frac{\{a\}^2(1 - \{a\})^2}{6a^2}, \tag{15}$$

where $\{a\}$ denotes the fractional part of a. The first term is a continuous contribution that grows with a, while the second term comes from the Bragg peaks inherited from the underlying lattice [12]. For $0 < a < 1$, where $\{a\} = a$, this simplifies to $\bar{\Lambda}(a) = (1 + a^2)/6$. At $a = 0$ the formula reduces to the integer lattice minimum $\bar{\Lambda} = 1/6$, and $\bar{\Lambda}$ increases monotonically with a.

Hyperuniformity can also be imposed directly in Fourier space. A stealthy hyperuniform pattern satisfies [13]

$$S(k) = 0 \quad \text{for all } |k| \leq K^*, \tag{16}$$

where K^* is a finite exclusion radius. The structure factor vanishes not just at $k = 0$ but over an entire interval, making these patterns invisible to scattering at long wavelengths. Stealthy patterns are realized as ground states of a pair potential that penalizes all density fluctuations at wavevectors $|k| \leq K^*$ [13]. They are characterized by the parameter $\chi = M(K^*)/[d(N-1)]$, where $M(K^*)$ is the number of independently constrained wavevectors satisfying $0 < |k| \leq K^*$ and d is the spatial dimension [13]. The ratio χ measures the fraction of degrees of freedom that are constrained to zero. Because $S(k) = 0$ over a finite interval rather than vanishing as a power law, stealthy patterns do not fit the standard three-class framework and α is not defined for them. However, $\bar{\Lambda}$ is still well-defined and is what we use to compare them with other patterns.

3 Methods

3.1 Pattern generation

Substitution tilings are generated by iteratively applying the substitution rules to a seed tile until the number of tiles exceeds 10^6, which is large enough that finite-size effects on $\sigma^2(R)$ are negligible. When higher precision is needed for spreadability-based extraction of α, we use larger chains with $N \sim 10^7$. The tile lengths are set by the eigenvalues of the substitution matrix so that the ratio of tile lengths is preserved under each substitution step, as described in Sec. 2.3. Points are placed at the left endpoint of each tile, producing a one-dimensional point pattern with N points and total domain length L.

URL patterns are generated by starting from an integer lattice of $N = 50{,}000$ points and displacing each site independently by a uniform random variable on $[-a/2, a/2]$, with a taking 17 values between 0.05 and 2.0 to span the range from weak to strong perturbations.

Data for stealthy configurations were provided by a collaborator, consisting of disordered ground states of the collective-coordinate potential [13] with $N = 2{,}000$ points at density $\rho = 1$. For each of three values $\chi \in \{0.1, 0.2, 0.3\}$, we use an ensemble of approximately 4,300 independent configurations, over which $\sigma^2(R)$ is averaged to obtain a representative ensemble estimate.

3.2 Extraction of α and $\bar{\Lambda}$

For substitution tilings, α is analytically determined from the eigenvalue formula (Eq. (11)). As an independent numerical check, we also measure α via diffusion spreadability.

To apply the spreadability method, each point pattern is embedded in a two-phase medium with packing fraction $\phi_2 = 0.35$ by centering a solid rod of half-length $a = \phi_2/(2\rho)$ at each point [10]. Extracting α from this medium requires the spectral density $\tilde{\chi}_V(k)$, which we obtain by first computing the structure factor $S(k)$ of the point pattern. We compute $S(k)$ by binning the point positions into a histogram with the number of bins chosen as the next power of 2 above $2L$ for FFT efficiency, and taking a real-valued fast Fourier transform. From $S(k)$, we get $\tilde{\chi}_V(k)$ using Eq. (12). In one dimension, the integral in Eq. (13) becomes the discrete sum

$$\mathcal{S}(\infty) - \mathcal{S}(t) = \frac{\Delta k}{\pi \phi_2} \sum_{n=1}^{N_k} \tilde{\chi}_V(k_n)\, e^{-k_n^2 D t}, \tag{17}$$

where $\Delta k = 2\pi/L$ is the wavevector spacing and $k_n = n\Delta k$. We evaluate this sum at 500 logarithmically spaced time points from $t = 1$ to $t = 10^6$ with $D = 1$. At short times, the spreadability is dominated by local structure and does not yet follow a power law, so we fit $\ln[\mathcal{S}(\infty) - \mathcal{S}(t)]$ vs. $\ln t$ over the time window $t \in [10^2, 10^5]$. Over this time frame, only the small-k behavior of $\tilde{\chi}_V(k)$ contributes, and from our fit we can extract α from the slope $= -(1+\alpha)/2$.

This simple log-slope approach can converge slowly for patterns whose subleading eigenvalue $|\lambda_2|$ is close to 1, because the correction terms in the spreadability asymptotics remain large even at $t \sim 10^5$–10^6. These corrections can be absorbed by fitting the excess spreadability to the form [14]

$$\ln[\mathcal{S}(\infty) - \mathcal{S}(t)] = -\frac{d+\alpha}{2} \ln t + \ln\left(\sum_{j=0}^{n} C_j\, t^{-j/2} \right), \tag{18}$$

where the $n+1$ coefficients C_j and the exponent α are determined by a nonlinear fit. This is the Type I fitting function of order n [14], the most general of three fitting forms proposed therein.

When $n = 0$, the sum reduces to a single constant and the fit is equivalent to the simple log-slope method. Each additional term absorbs one more order of subleading decay that would otherwise bias the extracted α at finite t. This expansion assumes that the spectral density $\tilde{\chi}_V(k)$ can be expanded in integer powers of k at the origin, which does not hold for quasicrystalline media with dense Bragg peaks [14]. For such media, the excess spreadability oscillates between two power-law envelopes $C_\pm\, t^{-(d+\alpha)/2}$ rather than decaying smoothly. Nevertheless, we find that the Type I fit still produces accurate results at moderate n.

$\bar{\Lambda}$ is extracted from the number variance $\sigma^2(R)$ in real space. We compute $\sigma^2(R)$ by randomly placing 30,000 windows of half-width R along the pattern with periodic boundary conditions and recording the variance of the point counts across all placements. We restrict $R < L/4$ to avoid finite-size artifacts and evaluate $\sigma^2(R)$ at 4,000 values of R, ranging from below the mean point spacing up to this maximum.

For Class I patterns, $\bar{\Lambda}$ is extracted as the mean of $\sigma^2(R)$ over the upper two-thirds of the sampled R range, where the variance has settled into its large-R asymptotic behavior. For quasicrystals, whose variance oscillates persistently at all scales due to the dense Bragg spectrum, this running average converges to a stable mean level that we take as $\bar{\Lambda}$.

For Class II and Class III patterns, $\bar{\Lambda}$ diverges and we instead extract $\Lambda_{\rm II}$ and $\Lambda_{\rm III}$ by fitting the large-R variance data to the scaling forms in Eqs. (8)–(9). For Class II we fit $\sigma^2(R) = \Lambda_{\rm II} \ln R + b$, where the additive constant b absorbs subleading contributions. For Class III we fit $\sigma^2(R) = A\, R^\beta$ with both A and β as free parameters, yielding $\Lambda_{\rm III} = A$ and $\alpha = 1 - \beta$. Leaving β free provides an independent determination of α from the variance data, which we compare against the eigenvalue-formula prediction.

3.3 Validation

We verify both the variance and spreadability pipelines against systems whose exact results are known analytically.

As a first check on the sliding-window variance algorithm, we generated a one-dimensional Poisson process with $N = 50{,}000$ points at unit density and computed $\sigma^2(R)$ using 30,000 random window placements at each of 50 values of R, averaged over 20 independent realizations. The measured variance recovers the exact scaling $\sigma^2(R) = 2\rho R$ to within 0.3% mean relative error across the full R-range (Fig. 9).

We then computed $\bar{\Lambda}$ for URL patterns at 17 values of the displacement amplitude $a \in [0.05, 2.0]$, each with $N = 50{,}000$ points. The numerical values agree with the exact formula (Eq. 15) [12] to better than 0.3% at every a tested, verifying that the running-average extraction of $\bar{\Lambda}$ is accurate for both weak and strong perturbations.

For the Fibonacci chain at $N \approx 1{,}000{,}000$ tiles with 30,000 window placements, our procedure yields $\bar{\Lambda} = 0.200$, which agrees with the $\bar{\Lambda} = 0.20110$ computed via a theta-function method [6] to within 0.55%. The small discrepancy is consistent with finite-size effects at our chain length.

As a check on the spreadability pipeline, we computed α for the Fibonacci chain ($N \approx 15{,}000{,}000$) using the log-slope fit over the window $t \in [10^2, 10^5]$ described in Sec. 3.2. The extracted value $\hat{\alpha} = 3.05$ agrees with the eigenvalue-formula prediction $\alpha = 3$ to within 2%. Higher precision ($\sim$0.01%) has been achieved using larger systems and thermodynamic-limit extrapolation [10], but our simpler log-slope fit on a single finite chain is sufficient for the classification purposes of this work.

4 Results

4.1 Metallic-mean family

We begin with the metallic-mean family $S \to L$, $L \to L^n S$, shown in Fig. 2 for $n = 1$ through 10. Table 1 lists the first five members of this family, spanning the Fibonacci, silver, bronze, copper, and nickel chains. As n grows, the long tile becomes much longer than the short tile: short tiles make up 38% of the Fibonacci sequence but only 9% at $n = 10$.

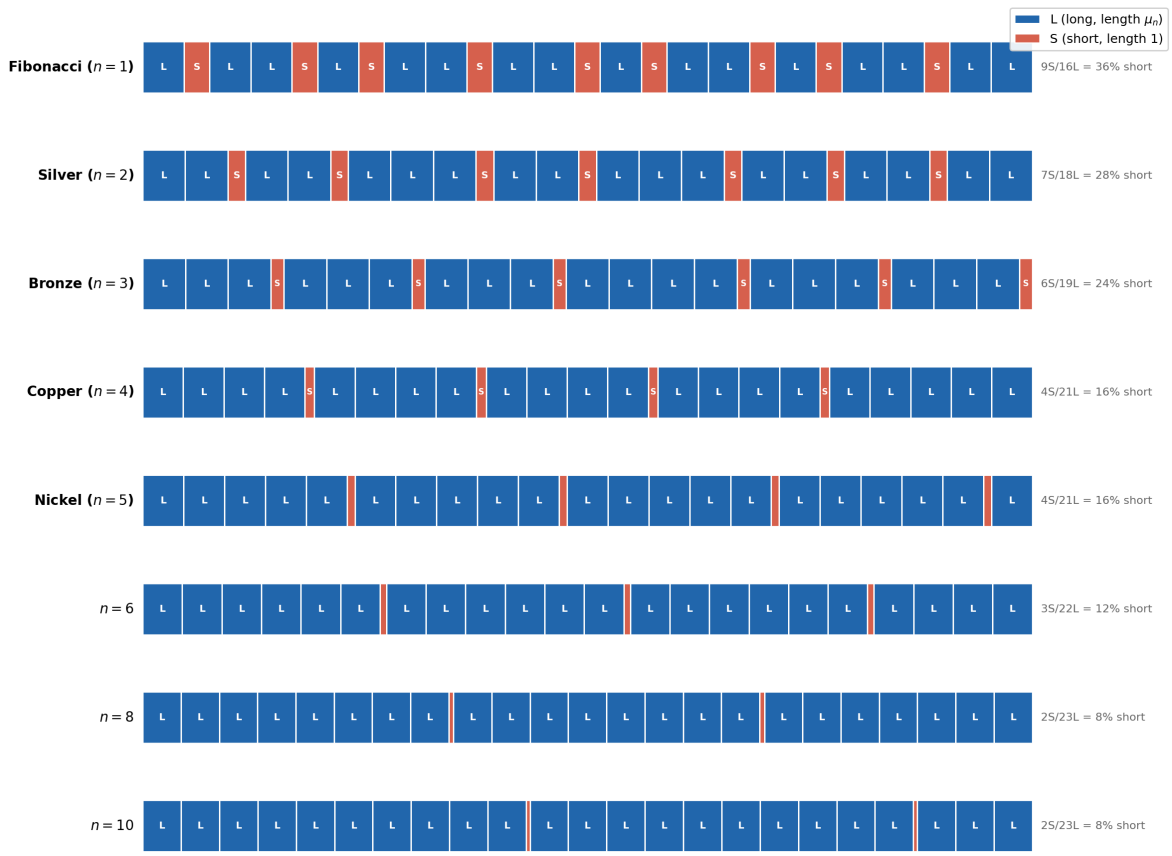

Figure 2: Metallic-mean tile sequences for selected n from 1 to 10, 25 tiles each. Long tiles (blue) have length μ_n, while short tiles (red) have length 1.

As shown in Sec. 2.3, the eigenvalue formula predicts $\alpha = 3$ for every member of this family regardless of n. Table 1 confirms this: the spreadability-based estimates of α agree with the prediction to within 5% for all five chains, with the largest errors at $n = 5$ and $n = 4$ due to log-periodic oscillations in $\mathcal{S}(t)$. The integer lattice ($\bar{\Lambda} = 1/6$) is included as the proven global minimum [4], and only the Fibonacci $\bar{\Lambda}$ was previously computed [6].

n	Chain	μ_n	ρ	α (eigenvalue)	α (spreadability)	$\bar{\Lambda}$
—	Integer lattice	—	1.000	∞	—	0.167
1	Fibonacci	1.618	0.724	3	3.05	0.200
2	Silver	2.414	0.500	3	2.99	0.250
3	Bronze	3.303	0.361	3	2.99	0.282
4	Copper	4.236	0.276	3	3.08	0.293
5	Nickel	5.193	0.222	3	2.88	0.310

Table 1: Metallic-mean chains for $n = 1$ through 5.

All five chains have $\alpha = 3$, despite having very different tile-length ratios and overall structures. This demonstrates that α is determined entirely by the eigenvalues of the substitution matrix, not by the geometry of the tiles. For each pattern, the variance is bounded and oscillates about a mean level $\bar{\Lambda}$ (Fig. 3). These $\bar{\Lambda}$ values are all different, increasing monotonically from 0.200 (Fibonacci) to 0.310 (Nickel), which shows that α and $\bar{\Lambda}$ indeed capture independent aspects of the pattern.

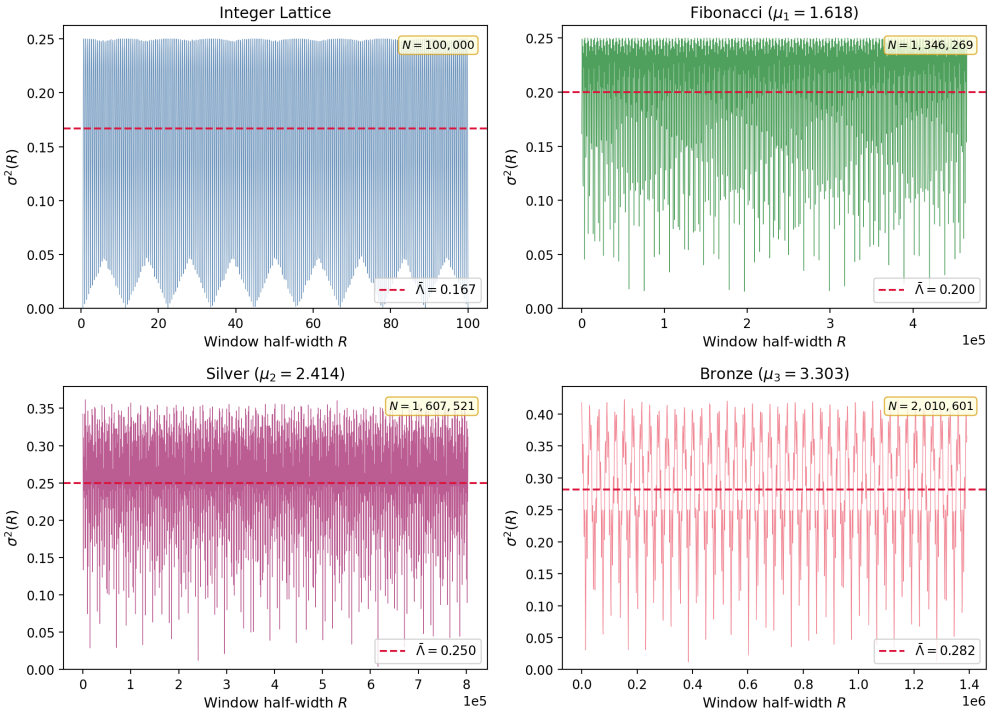

Figure 3: Bounded number variance $\sigma^2(R)$ for the integer lattice and three metallic-mean chains. All four patterns are Class I, but their mean variance levels ($\bar{\Lambda}$) differ. The three metallic-mean chains share $\alpha = 3$; the integer lattice has $\alpha = \infty$.

Extending the computation beyond $n = 5$, $\bar{\Lambda}$ continues to increase monotonically and appears to approach a limiting value near $1/3$ as $n \to \infty$, as shown in Fig. 4 below. Whether this limit is exactly $1/3$ remains a conjecture.

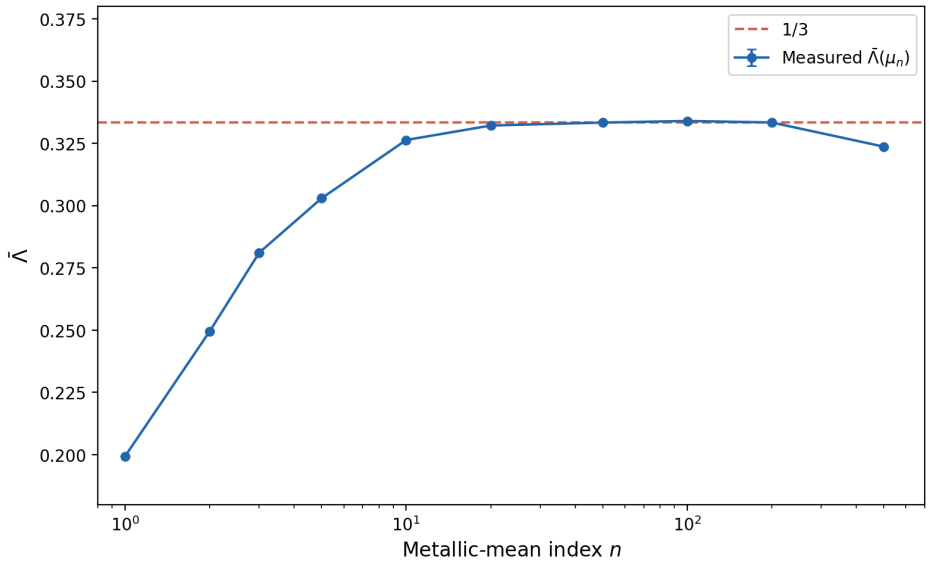

Figure 4: $\bar{\Lambda}$ for the metallic-mean family, computed for $n = 1$ through 500. The values increase toward a limit near 1/3 (dashed line), with some noise at large n due to reduced chain sizes.

4.2 Structural constraints on α

It is not a coincidence that metallic-mean chains give $\alpha = 3$. This result is forced by the structure of the substitution matrix. The eigenvalue formula (Eq. 11) combined with the condition $|\det M| = 1$ places an upper bound on α. For a rank-r Pisot matrix with $|\det M| = 1$, the product of all eigenvalues satisfies $|\lambda_1\lambda_2\cdots\lambda_r| = 1$. If all subleading eigenvalues have the same magnitude $|\lambda_i| = |\lambda_2|$ for $i \geq 2$, this gives $\lambda_1|\lambda_2|^{r-1} = 1$, so $|\lambda_2| = \lambda_1^{-1/(r-1)}$. Substituting into the eigenvalue formula yields the maximum

$$\alpha \leq 1 + \frac{2}{r-1}. \tag{19}$$

For rank 2, this gives $\alpha \leq 3$, which the metallic-mean chains saturate. For rank 3, $\alpha \leq 2$. Higher-rank matrices with $|\det M| = 1$ are even more constrained. To reach values of α other than 3, we can either increase the rank of the substitution matrix or relax the condition $|\det M| = 1$.

Increasing to rank 3 lowers the bound to $\alpha \leq 2$ and opens the range $1 < \alpha \leq 2$. For example, consider the substitution matrix with the three-letter rule $a \to aac$, $b \to ac$, $c \to b$ [15]

$$M_{\mathrm{BT}} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix},$$

with $|\det M| = 1$ and largest eigenvalue $\lambda_1 \approx 2.247$. The eigenvalue formula gives $\alpha = 1.545$, and we measure $\bar{\Lambda} = 0.377$ (Fig. 5). This chain also illustrates why the improved spreadability fitting (Eq. 18) is necessary for some patterns. Because $\alpha = 1.545$ is relatively small, the asymptotic power law $t^{-(1+\alpha)/2}$ takes longer to dominate over subleading corrections, and even at $t \sim 10^5$ the simple log-slope method gives $\hat{\alpha} = 2.28$, a 48% error. The improved Type I fit at order $n = 6$ absorbs these corrections and reduces the error to below 10% across all chain sizes tested, with the best result of $\hat{\alpha} = 1.57$ (1.7% error) at $N \approx 7 \times 10^6$ (Fig. 6).

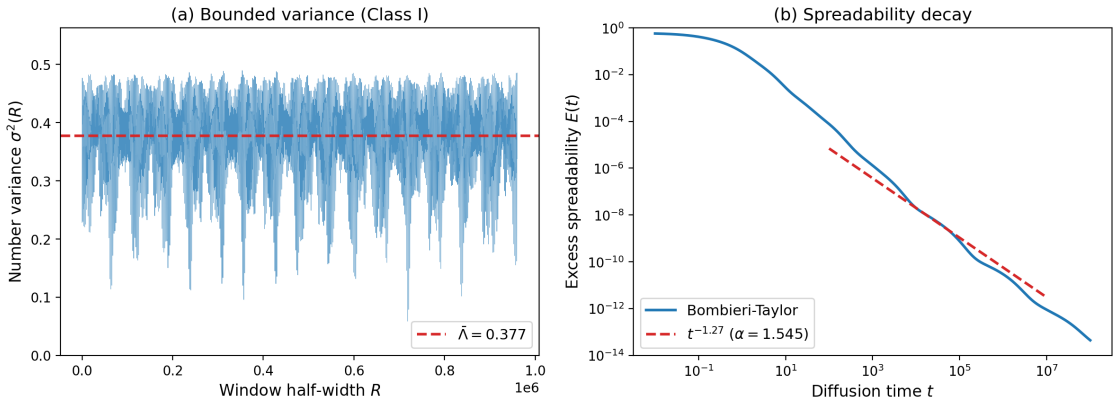

Figure 5: Bombieri-Taylor chain ($\alpha = 1.545$). (a) Bounded number variance with $\bar{\Lambda} = 0.377$. (b) Spreadability decay compared to the theoretical power law $t^{-1.27}$.

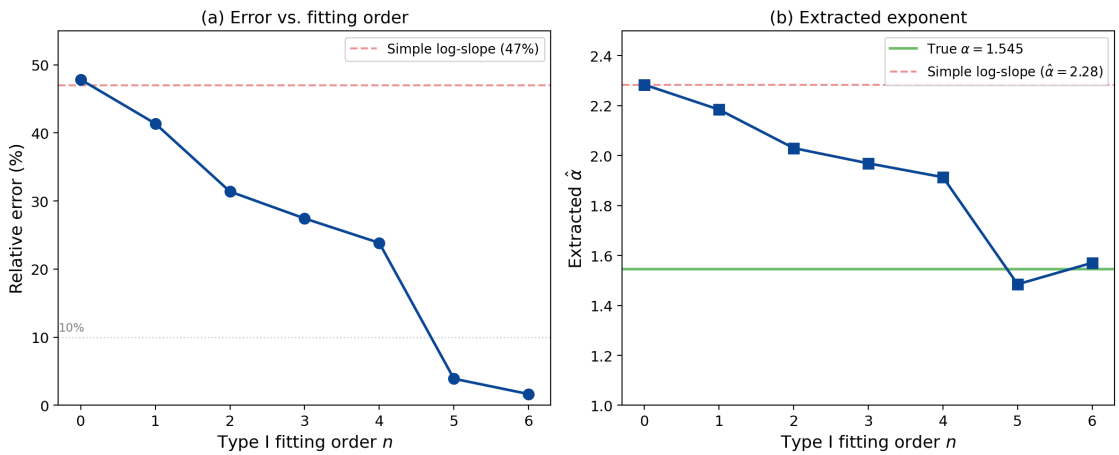

Figure 6: Improved spreadability fitting for the Bombieri-Taylor chain ($N = 6.8 \times 10^6$). (a) Relative error in the extracted α vs. Type I fitting order. The simple log-slope (red dashed) gives 48% error; each additional order absorbs subleading corrections, reaching 1.7% at order 6. (b) Extracted exponent $\hat{\alpha}$ converging from 2.28 (red dashed) toward the true value $\alpha = 1.545$ (green).

Among rank-3 matrices, a special case arises when λ_2 and λ_3 form a complex conjugate pair, so that $\det M = \lambda_1 |\lambda_2|^2$. Setting $|\det M| = 1$ forces $|\lambda_2| = 1/\sqrt{\lambda_1}$, which gives $\alpha = 2$ exactly, regardless of λ_1. In order to check this, we searched all 39,304 rank-3 matrices with non-negative integer entries and row sums ≤ 4. Of these, 504 are Pisot matrices with a complex eigenvalue pair, spanning 18 distinct values of λ_1 from 1.32 to 3.85, and all give $\alpha = 2$. One example, which we will later refer to as the cubic $\alpha = 2$ chain, is the rule $a \to bc$, $b \to c$, $c \to abc$, which gives $\bar{\Lambda} = 0.275$.

Because rank-2 matrices with $|\det M| = 1$ always give $\alpha = 3$ and rank ≥ 3 gives $\alpha \leq 2$, no Pisot matrix with $|\det M| = 1$ can produce α in the interval $(2, 3)$. This was confirmed by a further search over 24 million rank-4 matrices with row sums ≤ 4. However, we can construct matrices with α in $(2, 3)$ by allowing $|\det M| > 1$. For a rank-2 matrix, $|\det M| = \lambda_1 |\lambda_2|$, so $|\lambda_2| = |\det M|/\lambda_1$. Substituting into Eq. (11) gives

$$\alpha = 3 - \frac{2 \ln |\det M|}{\ln \lambda_1}, \tag{20}$$

which reduces to $\alpha = 3$ when $|\det M| = 1$ but yields $\alpha < 3$ when $|\det M| > 1$. For α to remain above 2, we need $|\det M| < \sqrt{\lambda_1}$. Because $|\det M|$ is a positive integer, the smallest useful value is $|\det M| = 2$, which requires $\lambda_1 > 4$. The simplest such matrix is $M = \left(\begin{smallmatrix} 0 & 2 \\ 1 & 4 \end{smallmatrix}\right)$, with rules $a \to b$, $b \to aabbbb$, $\lambda_1 = 2 + \sqrt{6} \approx 4.449$, giving $\alpha = 2.071$ (Fig. 7). A systematic search over all rank-2 matrices with row sums ≤ 10 found 138 matrices with $|\det M| = 2$ and α between 2.071 and 2.392, spanning 12 distinct α values. Table 2 lists all 12 groups. Within each group, $\bar{\Lambda}$ varies by factors of 5–15, showing that even matrices with identical eigenvalues can produce very different levels of density fluctuation.

α	Chains	$\bar{\Lambda}$ range
2.071	12	[0.32, 2.01]
2.087	10	[0.34, 1.57]
2.175	18	[0.32, 3.36]
2.199	14	[0.34, 3.57]
2.248	10	[0.33, 2.82]
2.271	16	[0.34, 3.76]
2.301	18	[0.33, 4.05]
2.323	4	[0.34, 2.99]
2.343	12	[0.33, 4.79]
2.362	12	[0.34, 4.80]
2.376	8	[0.33, 4.97]
2.392	4	[0.34, 5.08]

Table 2: Rank-2 matrices with $|\det M| = 2$ and $\alpha \in (2, 3)$. 138 chains total across 12 distinct α values.

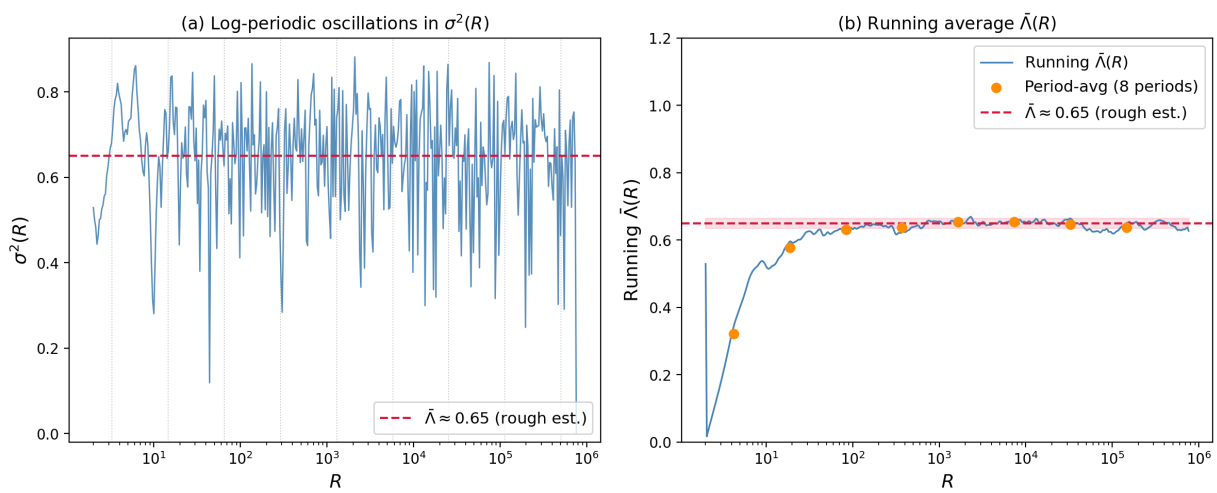

Figure 7: The simplest $|\det M| = 2$ chain ($a \to b$, $b \to aabbbb$, $\alpha = 2.071$, $N \approx 900{,}000$). (a) Number variance $\sigma^2(R)$ showing large log-periodic oscillations. (b) Running average $\bar{\Lambda}(R)$ converging toward $\bar{\Lambda} \approx 0.65$.

4.3 Classes II and III

The patterns discussed so far are all Class I ($\alpha > 1$). To complete the picture, we also examine two substitution tilings that fall into the weaker hyperuniformity classes.

The period-doubling chain, with rule $S \to L$, $L \to LSS$ and matrix $M = \left(\begin{smallmatrix} 0 & 2 \\ 1 & 1 \end{smallmatrix}\right)$, has eigenvalues $\lambda_1 = 2$ and $\lambda_2 = -1$. Because $|\lambda_2| = 1$, the eigenvalue formula gives $\alpha = 1$ exactly, placing it at the boundary between Class I and Class II. Its variance grows logarithmically, and fitting $\sigma^2(R) = \Lambda_{\mathrm{II}} \ln R + b$ yields $\Lambda_{\mathrm{II}} = 0.080$.

The 0222 chain, with rule $S \to LL$, $L \to SSLL$ and matrix $M = \left(\begin{smallmatrix} 0 & 2 \\ 2 & 2 \end{smallmatrix}\right)$, has $\lambda_1 = 1+\sqrt{5} \approx 3.236$ and $|\lambda_2| = \sqrt{5} - 1 \approx 1.236$. Because $|\lambda_2| > 1$, this is not a Pisot substitution, and the eigenvalue formula gives $\alpha = 0.639$, placing it in Class III. Its variance grows as a power law, and the free-parameter fit $\sigma^2(R) = A\,R^{\beta}$ yields $\Lambda_{\mathrm{III}} = 0.209$ with $\alpha = 1 - \beta = 0.684$. This is within 7% of the eigenvalue prediction $\alpha = 0.639$, which can be attributed to the difficulty of extracting a clean power-law exponent from finite variance data with oscillations.

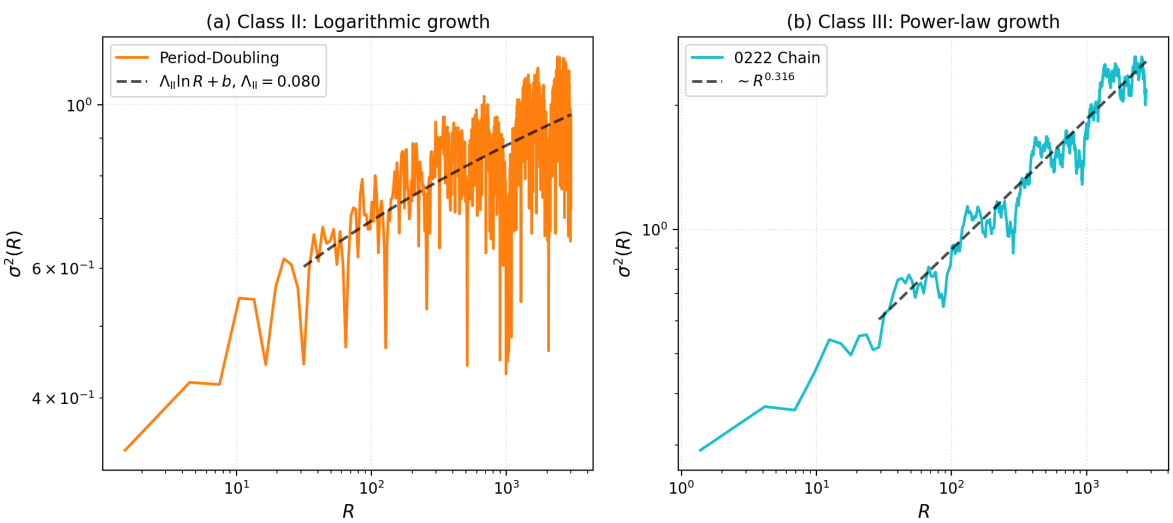

Figure 8: Number variance $\sigma^2(R)$ for Classes II and III. (a) Period-Doubling (Class II, $\alpha = 1$), with logarithmic growth and fitted coefficient $\Lambda_{\mathrm{II}} = 0.080$. (b) 0222 chain (Class III, $\alpha = 0.639$), with power-law growth $\sigma^2 \sim R^{0.316}$. Dashed lines show the fits described in Sec. 3.2.

4.4 Ranking Class I patterns by $\bar{\Lambda}$

Table 3 collects all Class I patterns studied in this work, sorted by $\bar{\Lambda}$, with the Class II and III results listed separately.

Pattern	Class	α	$\bar{\Lambda}$
Integer lattice	I	—	$1/6 = 0.167$
Fibonacci ($n = 1$)	I	3	0.200
Silver ($n = 2$)	I	3	0.250
Cubic $\alpha = 2$	I	2	0.275
Bronze ($n = 3$)	I	3	0.282
Copper ($n = 4$)	I	3	0.293
Nickel ($n = 5$)	I	3	0.310
URL ($a = 1$)	I	2	$1/3 = 0.333$
Stealthy $\chi = 0.3$	I	—	0.357
Bombieri-Taylor	I	1.545	0.377
Stealthy $\chi = 0.2$	I	—	0.526
Stealthy $\chi = 0.1$	I	—	1.021
Period-Doubling	II	1	$\Lambda_{\mathrm{II}} = 0.080$
0222 chain	III	0.639	$\Lambda_{\mathrm{III}} = 0.209$

Table 3: All patterns studied in this work, sorted by $\bar{\Lambda}$. Stealthy patterns have $S(k) = 0$ over a finite interval instead of vanishing as a power law, so α is not defined. Period-Doubling and 0222 chain have divergent $\bar{\Lambda}$ and are characterized by Λ_{II} and Λ_{III} instead.

For the URL, our numerical $\bar{\Lambda}$ values confirm the exact formula (Eq. 15) [12] to better than 0.3% across all 17 values of a tested[2]. At integer values of a, the Bragg contribution vanishes and $\bar{\Lambda} = a/3$. For stealthy configurations, $S(k) = 0$ over the entire interval $|k| \leq K^*$, and $\bar{\Lambda}$ decreases as χ increases, from 1.021 at $\chi = 0.1$ to 0.357 at $\chi = 0.3$. We find that $\bar{\Lambda} \approx 1/(\pi^2\chi)$ in the disordered regime, though this is an empirical approximation rather than an exact result. Constraining a larger fraction of Fourier modes to zero leaves less room for density fluctuations, driving $\bar{\Lambda}$ toward the lattice minimum.

Several features of the catalog stand out. As discussed in Sec. 4.1, the metallic-mean $\bar{\Lambda}$ appears to approach 1/3 as $n \to \infty$. This is the same value achieved by the URL at $a = 1$, a disordered pattern with continuous $S(k)$ and $\alpha = 2$. Whether this numerical coincidence reflects a deeper connection between the two systems or is accidental remains an open question.

The cubic $\alpha = 2$ chain ($\bar{\Lambda} = 0.275$) is a deterministic substitution tiling whose $\bar{\Lambda}$ matches that of a URL with $a \approx 0.8$. These two patterns share the same exponent and comparable fluctuation amplitude, but achieve hyperuniformity through completely different mechanisms. Similarly, stealthy patterns at $\chi = 0.3$ have $\bar{\Lambda} = 0.357$, which exceeds every metallic-mean chain, despite the fact that stealthy patterns suppress $S(k)$ to zero over a finite interval. Thus, perfect small-k suppression does not guarantee tight local spatial order.

The 138 chains with $|\det M| = 2$ (Table 2) provide the strongest evidence that α and $\bar{\Lambda}$ are independent. Within each α group, $\bar{\Lambda}$ varies by factors of 5 to 15, depending on the full substitution rule rather than the eigenvalues alone.

[2] Shown in Fig. 10

5 Conclusion

The standard three-class scheme for hyperuniformity sorts patterns by how fast $S(k)$ vanishes near the origin, but it does not distinguish between patterns within the same class. In this work, we computed both α and $\bar{\Lambda}$ for a broad collection of one-dimensional hyperuniform point patterns and found that these two metrics capture complementary information about large-scale structure. Patterns sharing the same α can differ in $\bar{\Lambda}$ by more than an order of magnitude, and patterns with different α can converge to the same $\bar{\Lambda}$. For the metallic-mean family, $\bar{\Lambda}$ increases monotonically with n and appears to approach $1/3$, the value achieved by a uniformly randomized lattice at $a = 1$.

The algebraic structure of the substitution matrix constrains which values of α are accessible. For Pisot matrices with $|\det M| = 1$, the bound $\alpha \leq 1+2/(r-1)$ locks rank-2 matrices at $\alpha = 3$ and rank-3 matrices at $\alpha \leq 2$, leaving the interval $(2, 3)$ empty. Relaxing to $|\det M| = 2$ opens this gap, with 138 matrices found producing α values between 2.07 and 2.39. Within each α group, $\bar{\Lambda}$ varies widely, confirming that the full substitution rule, not just the eigenvalue spectrum, determines the amplitude of density fluctuations.

From a physical standpoint, α controls the long-time decay rate of diffusion spreadability through the corresponding two-phase medium, while $\bar{\Lambda}$ sets the amplitude of the remaining fluctuations. Two patterns with the same α but different $\bar{\Lambda}$ will have the same asymptotic spreadability decay rate, but differ in the density fluctuations that persist at any finite scale. Similarly, stealthy patterns suppress $S(k)$ to zero over a finite interval, but their surface-area coefficient $\bar{\Lambda}$ can exceed that of every metallic-mean chain, showing that perfect small-k suppression does not guarantee tight local spatial order.

One limitation of the catalog is that the only $\alpha = 3$ patterns available are the ordered metallic-mean chains. A disordered family at the same exponent would allow a direct test of how order and disorder affect $\bar{\Lambda}$ at fixed α, but no such family is currently known in one dimension. Several other questions remain open, including whether an analytical formula exists for the metallic-mean $\bar{\Lambda}$, whether the sequence converges to exactly $1/3$, and whether the two-axis framework extends naturally to two-dimensional quasicrystals such as Penrose and Ammann-Beenker tilings.

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Appendix A: Additional figures

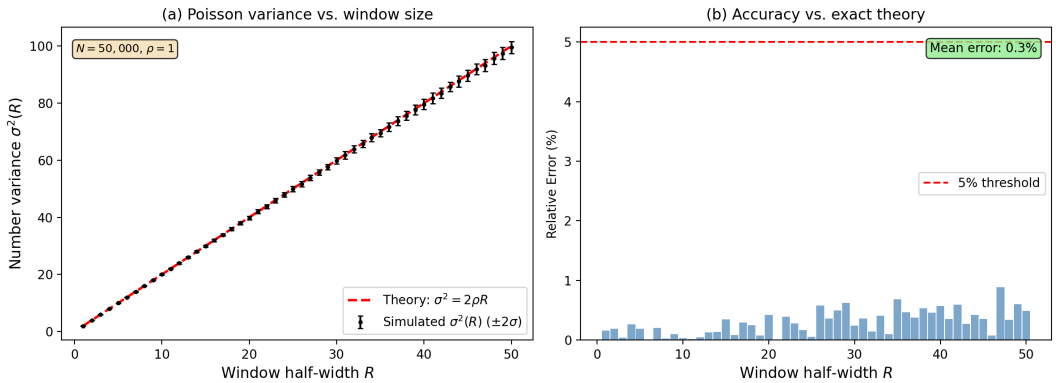

Figure 9: Poisson process variance benchmark. The measured $\sigma^2(R)$ agrees with the theoretical scaling $\sigma^2 = 2\rho R$ (dashed) to within 0.3% mean relative error, averaged over 20 independent realizations.

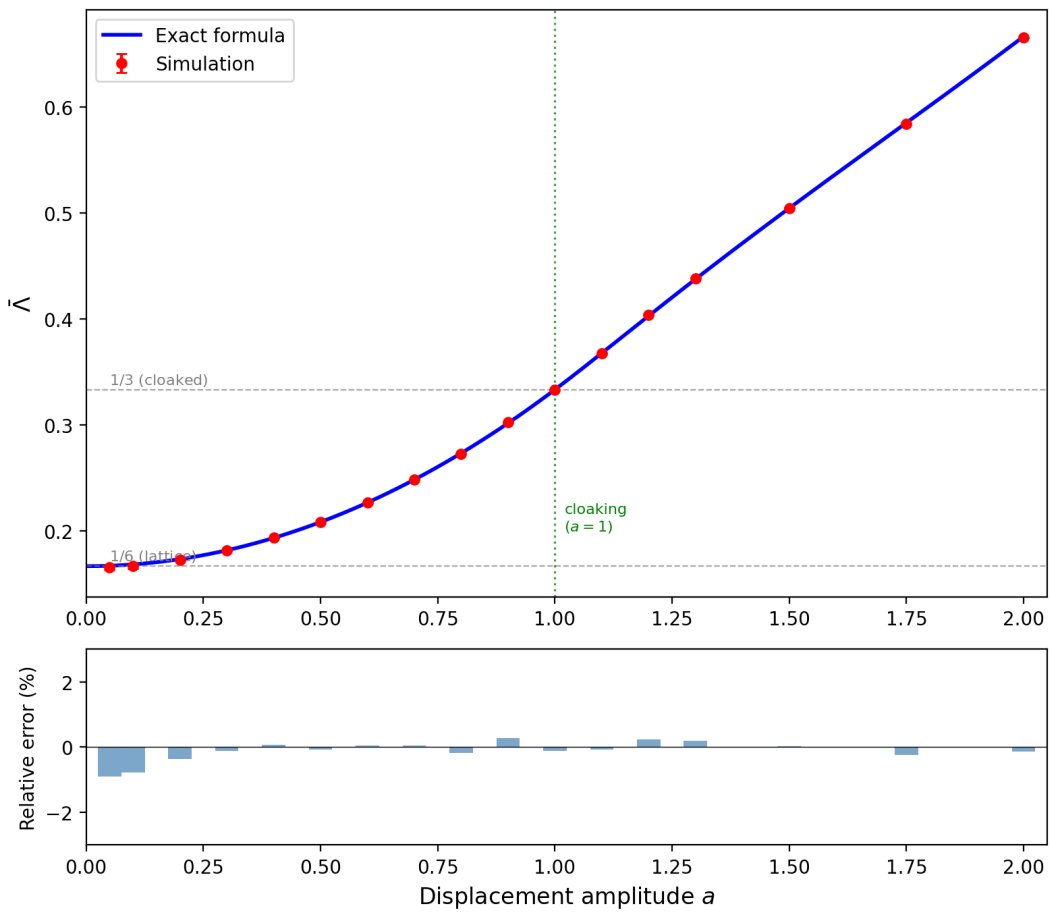

Figure 10: Numerical $\bar{\Lambda}(a)$ vs. the exact formula (Eq. 15) for 17 values of a for URLs.